

# Math 114 Class notes Fall 2026

To accompany  
*Precalculus 2e*  
by *Abramson*

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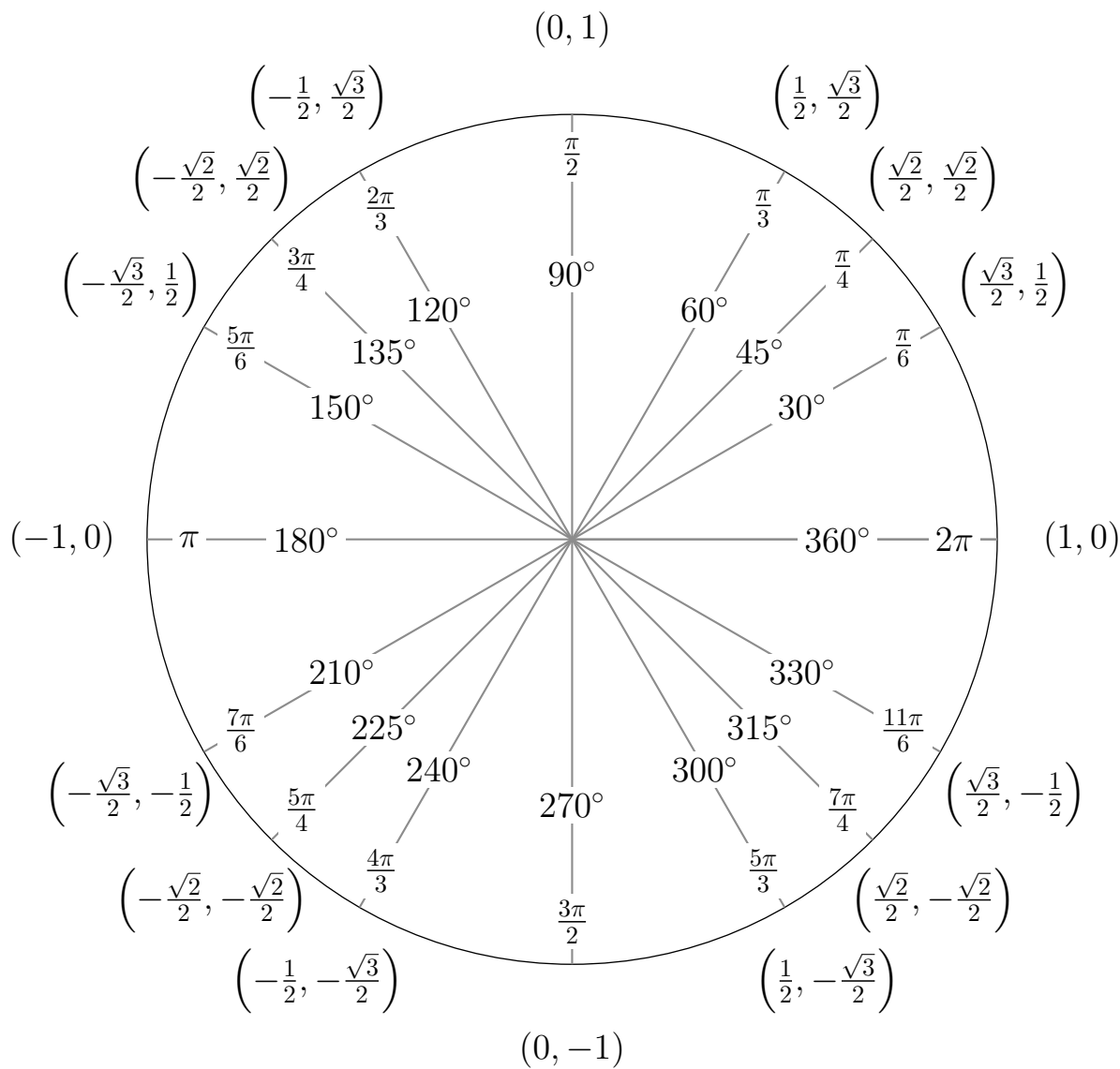
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# Math 114 Formula Sheet

# Unit Circle

$$(\cos(\theta), \sin(\theta))$$



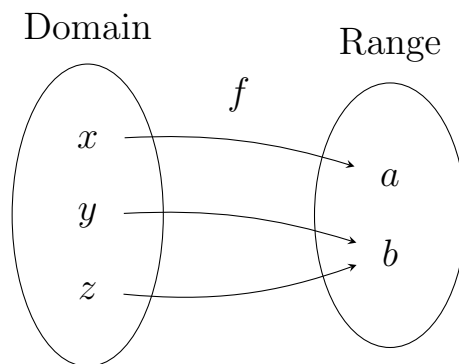
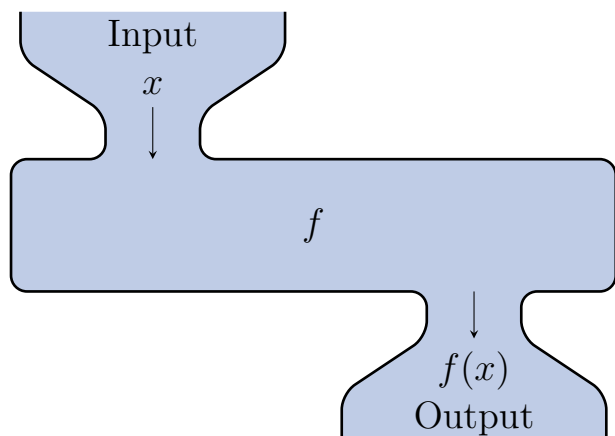
$$180^\circ = \pi \text{ rad}$$

## 1.1: Functions and Function Notation

### Definition.

A **function** is a relation in which each possible input value leads to exactly one output value.

The input values make up the **domain**, and the output values make up the **range**.



**Example.** Let  $f(x) = 2x^2 - 2x + 1$ . Evaluate the following

$$f(1)$$

$$f(-2)$$

$$f(a)$$

$$f(a + h)$$

$$\frac{f(a + h) - f(a)}{h}$$

**Example.** Suppose that the following tables represent a carnival truck’s menu prices. Which table would represent a valid function?

| Item           | Price |
|----------------|-------|
| Funnel cake    | \$8   |
| Chicken strips | \$9   |
| Ribbon fries   | \$7   |
| Cheese nachos  | \$6   |

| Item           | Price |
|----------------|-------|
| Funnel cake    | \$8   |
| Chicken strips | \$8   |
| Ribbon fries   | \$7   |
| Cheese nachos  | \$6   |

| Item           | Price |
|----------------|-------|
| Funnel cake    | \$8   |
| Chicken strips | \$8   |
| Ribbon fries   | \$8   |
| Funnel cake    | \$6   |

**Example.** Based on an old match-makers rule, the socially “acceptable” minimum age of someone you can date is half your age plus seven. Make a function that takes your age,  $x$ , and finds the minimum age of someone you can date,  $y$ .

**Example.** Create a table to represent a function that takes each month and returns the number of days (in a non-leap year).

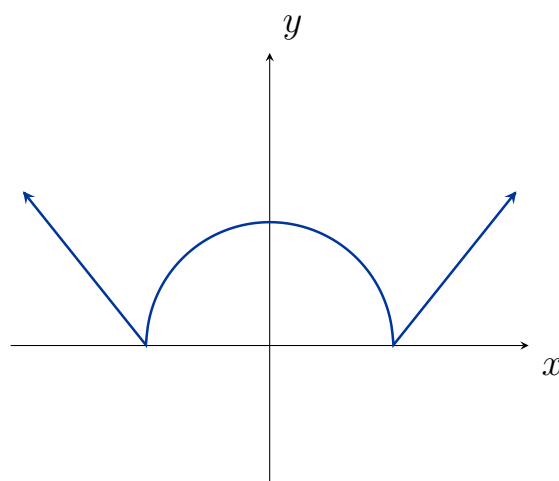
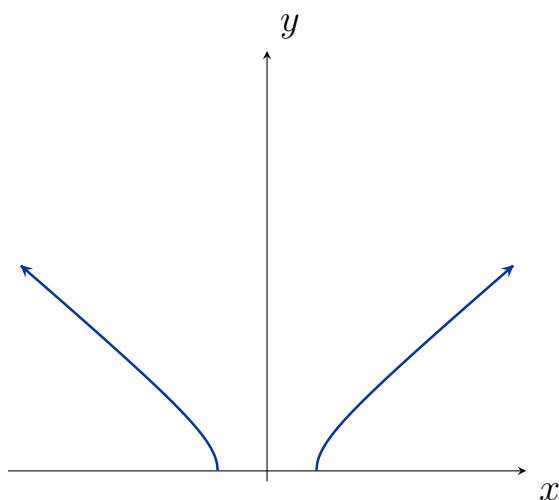
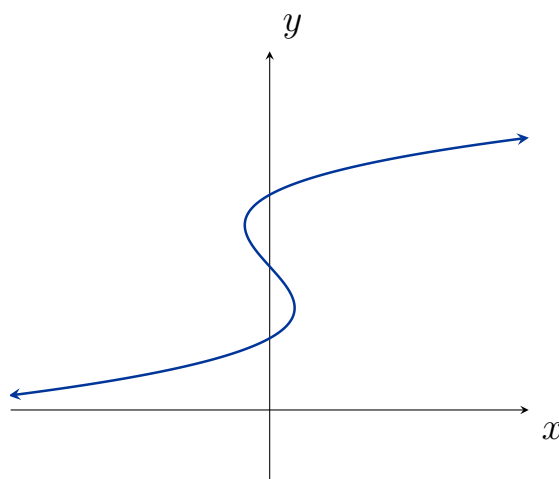
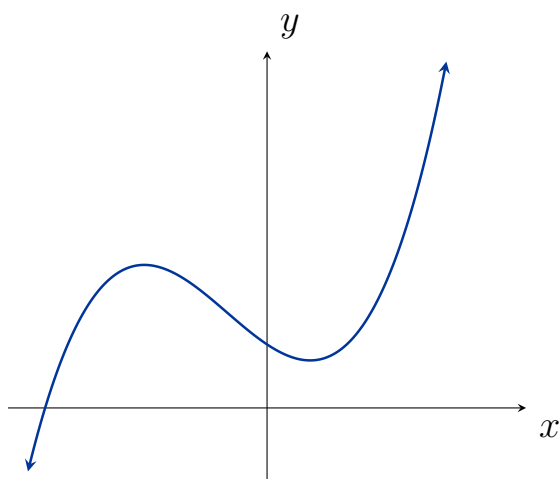
**Example.** If possible, express the relationship  $2n + 6p = 12$  as a function  $p = f(n)$ . Graph the result.

**Example.** If possible, express the relationship  $x^2 + y^2 = 1$  as a function  $y = f(x)$ . Graph the result.

**Definition. (Vertical Line Test)**

A curve in the  $xy$ -plane is the graph of a function  $y = f(x)$  (an explicit function) if and only if each vertical line intersects it in at most one point

**Example.** Use the vertical line test on the following graphs to determine which graphs may represent an explicit function:





**Definition. (One-to-One)**

The **horizontal line test** determines if every horizontal line passes through the graph of a function in at most one point.

A function is said to be **one-to-one** if each output value corresponds to *exactly* one input value (and hence passes the horizontal line test).

**Example.** Which of the following represent a one-to-one function?

[Graph](#)

$$y = x^2$$

$$y^2 = x$$

$$f(x) = \sqrt{x}$$

$$y = |x|$$

$$f(x) = (x + 1)(x - 1)(x - 2)$$

The equation of a circle:  
 $x^2 + y^2 = 1$

The area of a circle:

$$A = \pi r^2$$

[Graphs of basic functions](#)

## 1.2: Domain and Range

### Definition. (Interval notation)

We can specify the domain and/or range of a function using interval notation:

- Strict inequalities (exclusive) are represented using parenthesis:

$$x > 3 \iff (3, \infty) \qquad x < 9 \iff (-\infty, 9)$$

*Note:* Parenthesis are **always** used when an interval's end points are infinite

- Nonstrict inequalities (inclusive) are represented using square brackets:

$$7 \leq x \leq 11 \iff [7, 11]$$

- Disjoint intervals can be unioned together:

$$x \neq -2 \iff (-\infty, -2) \cup (-2, \infty)$$

- Any combination of the above can be used:

$$[-1, 1] \cup (2, 3] \cup (4, 5) \cup [6, \infty)$$



Recall that the domain is the set of all *possible* input values.

**Example.** Find the domain of the following functions (use interval notation): [Graphs](#)

$$f(x) = x - 3$$

$$A = \pi r^2$$

$$y = \sqrt{x - 1}$$

$$y = \frac{x + 2}{x^2 - 4}$$

$$y = \sqrt[3]{x - 4}$$

$$f(x) = \frac{\sqrt[4]{x^2 + 4}}{(x + 1)\sqrt{x - 1}}$$

**Example.** Identify the domain and range of the following:

[Graphs](#)

$$x(x+1)(x+2)^2, \quad x \geq -2$$

$$\sin(x)$$

$$\log(x)$$

$$2^x, \quad x \geq -1$$

**Definition.**

A **piecewise** function is a function with different definitions for different portions of the domain.

**Example.** Rewrite the following as piecewise functions:

$$|x| = \qquad \qquad \qquad \frac{x}{|x|} =$$

$$|x - 1| + |4 - x| =$$

**Example.** Graph the following piecewise function:

[Graphs](#)

$$f(x) = \begin{cases} x^2, & \text{if } x \leq 1 \\ 3, & \text{if } 1 < x \leq 2 \\ x, & \text{if } x > 2 \end{cases}$$

### 1.3: Rates of Change and Behavior of Graphs

**Definition. (Rate of Change)**

A rate of change describes how an output quantity changes relative to the change in the input quantity. The units on a rate of change are “output units per input units”. The **average rate of change** between two input values is the total change of the function values divided by the input values.

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

**Example.** Suppose the following table contains temperatures recorded through the day.

| $h$    | 7 AM         | 9 AM         | 11 AM        | 1 PM         | 3 PM         |
|--------|--------------|--------------|--------------|--------------|--------------|
| $T(h)$ | $78^\circ F$ | $81^\circ F$ | $87^\circ F$ | $93^\circ F$ | $98^\circ F$ |

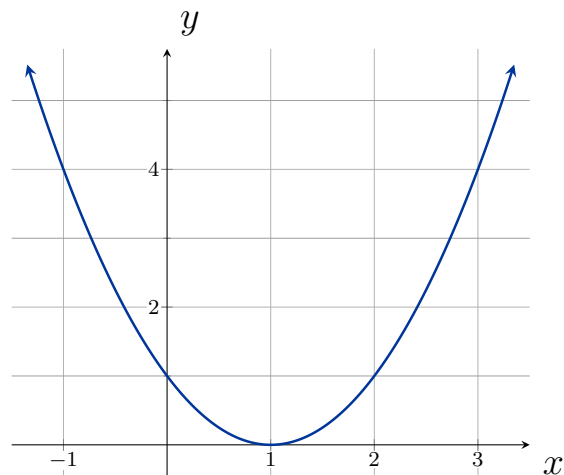
Find the average rate of change between

7 AM and 9 AM

9 AM and 11 AM

7 AM and 3 PM

**Example.** Using the following graph, determine the average rate of change on the interval  $[-1, 2]$



**Example.** Find the average rate of change of  $f(x) = x^2 - \frac{1}{x}$  on the interval  $[2, 4]$ .



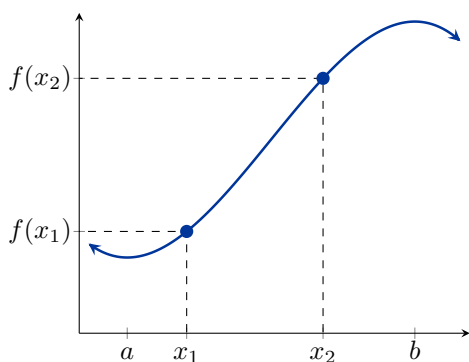
**Example.** Find the expression for the average rate of change of  $g(t) = t^2 + 3t + 1$  on the interval  $[0, a]$ .

**Example.** Find the expression for the average rate of change of  $j(t) = \frac{1}{t+3}$  on the interval  $[1, 1+h]$ .

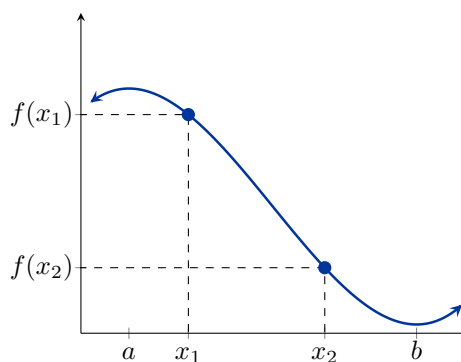
**Definition.**

Consider the function  $f(x)$  on the interval  $(a, b)$ . Given *any* two numbers  $x_1$  and  $x_2$  in  $(a, b)$  where  $x_1 < x_2$ , we say  $f$  is

increasing if  $f(x_1) < f(x_2)$



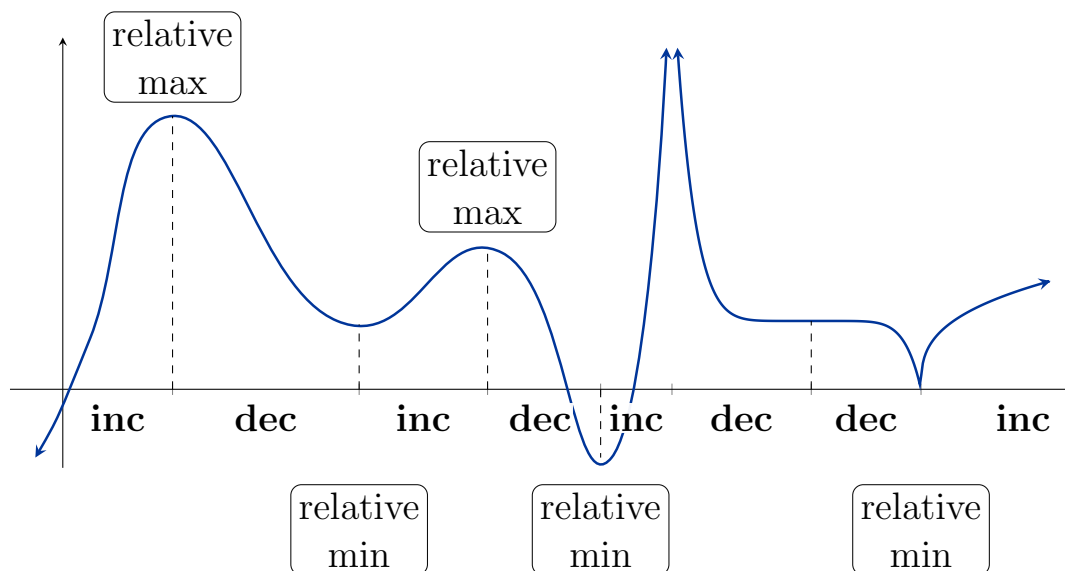
decreasing if  $f(x_1) > f(x_2)$



**Definition. (Relative/Local Extrema)**

A function  $f$  has a

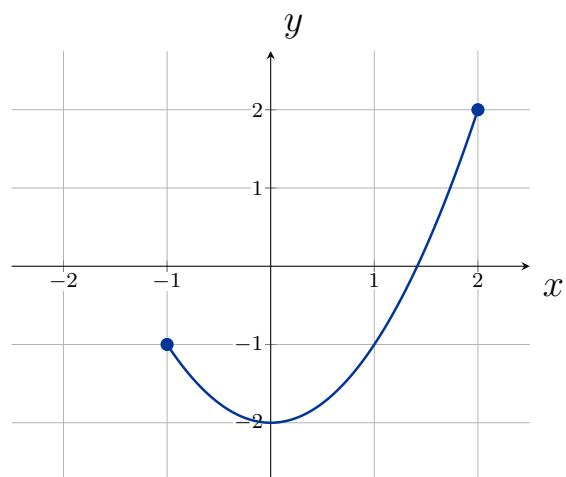
- **relative maximum** at  $x = c$  if  $f(c) \geq f(x)$  for every  $x$  in  $(a, b)$
- **relative minimum** at  $x = c$  if  $f(c) \leq f(x)$  for every  $x$  in  $(a, b)$

**Definition. (Absolute Extrema)**

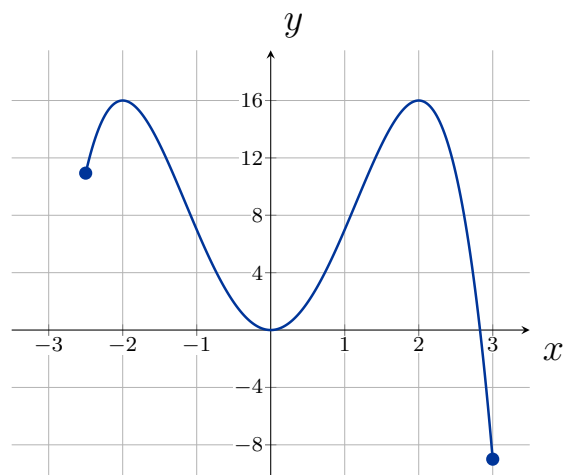
A function  $f$  defined on a domain  $D$  has an

- **absolute maximum** at  $x = c$  if  $f(c) \geq f(x)$  for every  $x$  in  $D$
- **absolute minimum** at  $x = c$  if  $f(c) \leq f(x)$  for every  $x$  in  $D$

**Example.** Using the following graph, find any relative and absolute extrema.



**Example.** Using the following graph, find any relative and absolute extrema.



## 1.4: Composition of Functions

### Definition. (The Composition of Two Functions)

Let  $f$  and  $g$  be functions. Then the composition of  $g$  and  $f$  is the function  $g \circ f$  defined by

$$(g \circ f)(x) = g(f(x))$$

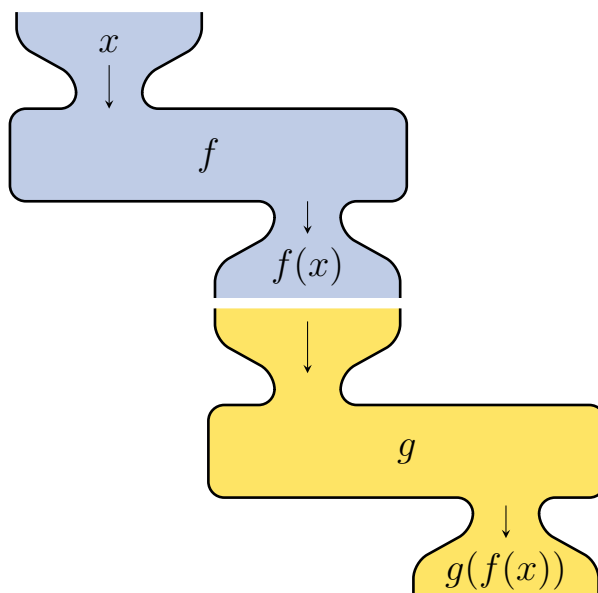
The domain of  $g \circ f$  is the set of all  $x$  in the domain of  $f$  such that  $f(x)$  lies in the domain of  $g$ .

**Example.** Let  $f(x) = \sqrt{x+1}$  and  $g(x) = 4 - x$ . Find the domain of the following:

$$g(f(x)) =$$

$$f(g(x)) =$$

$$f(f(x)) =$$



**Example.** For the following, find functions  $g(x)$  and  $h(x)$  so that  $f(x) = g(h(x))$ :

$$f(x) = \frac{1}{x+2}$$

$$f(x) = \sqrt{x-4}$$

$$f(x) = (x^2 + 3x - 8)^4$$

$$f(x) = \frac{x-3}{\sqrt{-(x-3)}}$$